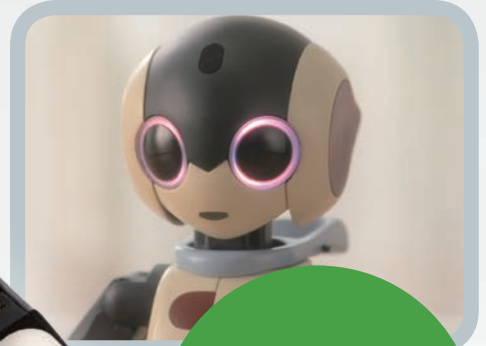
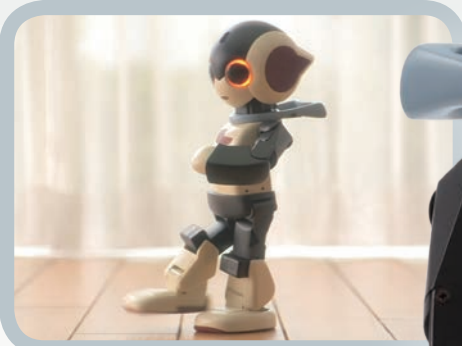
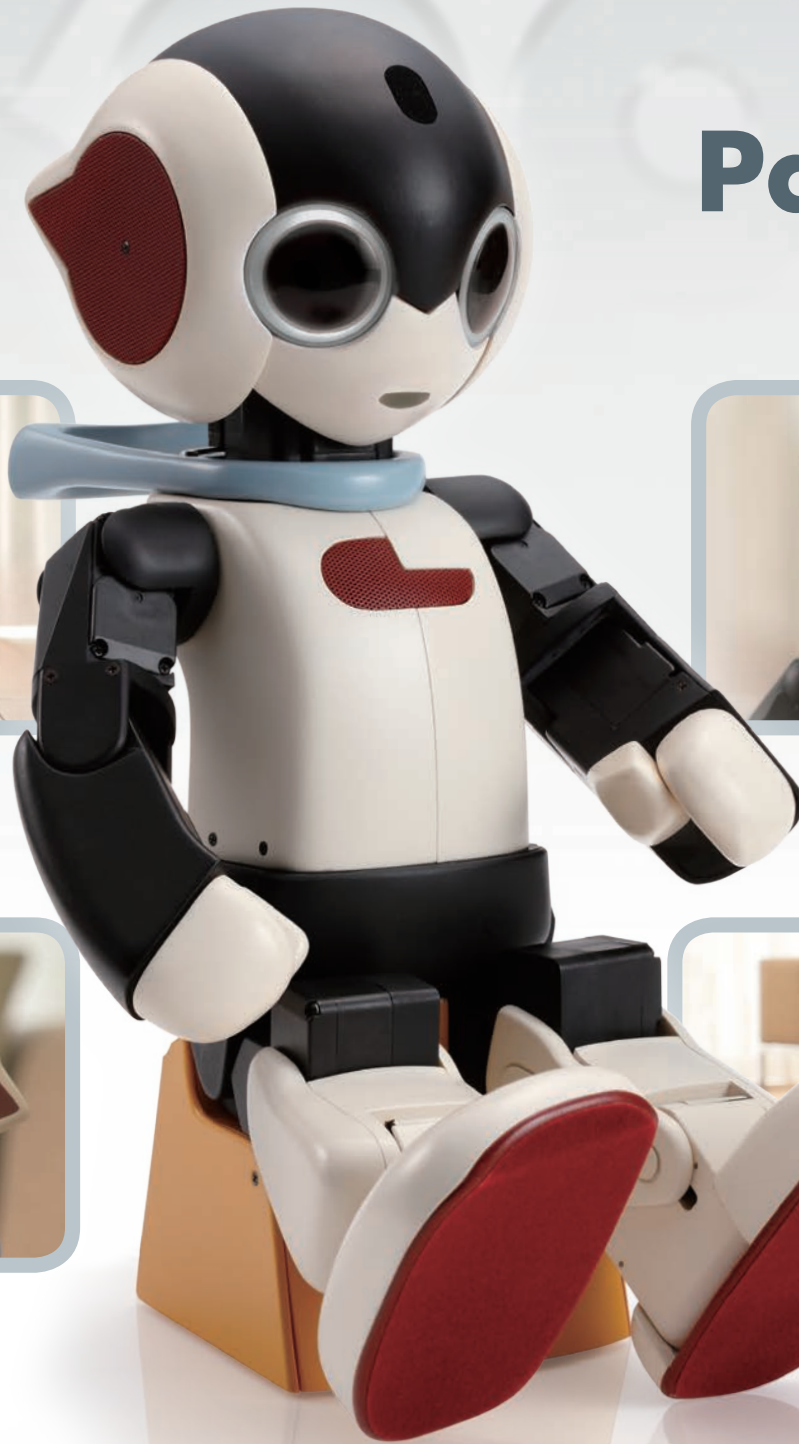
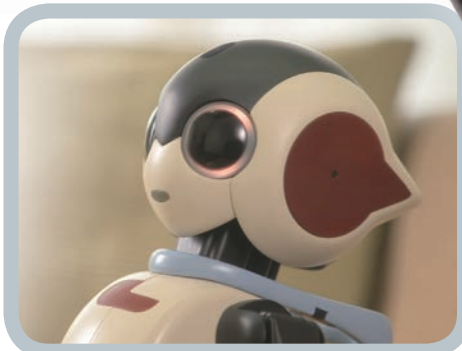


Build your own Robi

Pack **13**



Stages 47-50:
Continue building
up Robi's left leg



Build your own **Robi**

CONTENTS

Assembly Guide **p241**

Stage 47: Prepare Robi's left knee servo **p241**

Stage 48: Assemble Robi's left knee then test his left leg **p245**

Stage 49: Build up and test Robi's left leg **p253**

Stage 50: Prepare to join Robi's legs for the first time **p260**

DeAGOSTINI
MODEL SPACE™
www.model-space.com

NOT SUITABLE FOR CHILDREN UNDER THE AGE OF 14.
THIS PRODUCT IS NOT A TOY AND IS NOT DESIGNED FOR USE IN
PLAY. ITEMS MAY VARY FROM THOSE SHOWN. THERE MAY BE
COLOUR VARIATIONS OWING TO THE MATERIALS USED.

All rights reserved © 2016

Published in the UK by De Agostini UK Ltd,
Battersea Studios 2, 82 Silverthorne Road,
London SW8 3HE

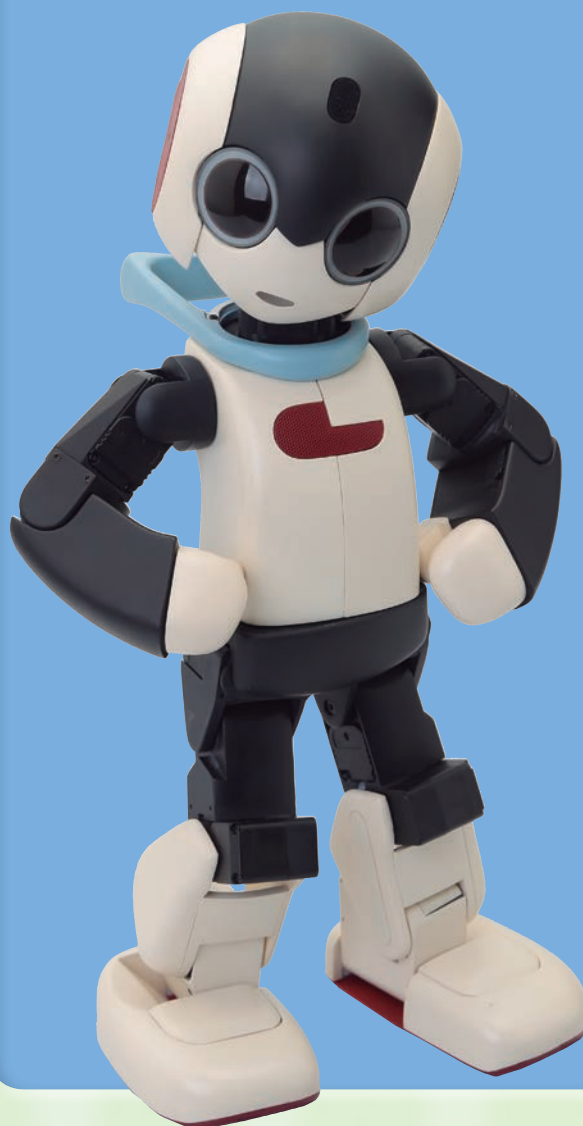
Published in the USA by De Agostini Publishing
USA, Inc., 121 E. Calhoun Street
Woodstock, IL 60098

Packaged by Continuo Creative,
39-41 North Road, London N7 9DP

Printed in EU

The names and likenesses of all Robi characters,
artwork and distinctive lettering are protected
under the copyright and trademark laws of the
United States and all foreign countries.

PICTURE CREDITS:
p352-3, all: Unplugged Inc.

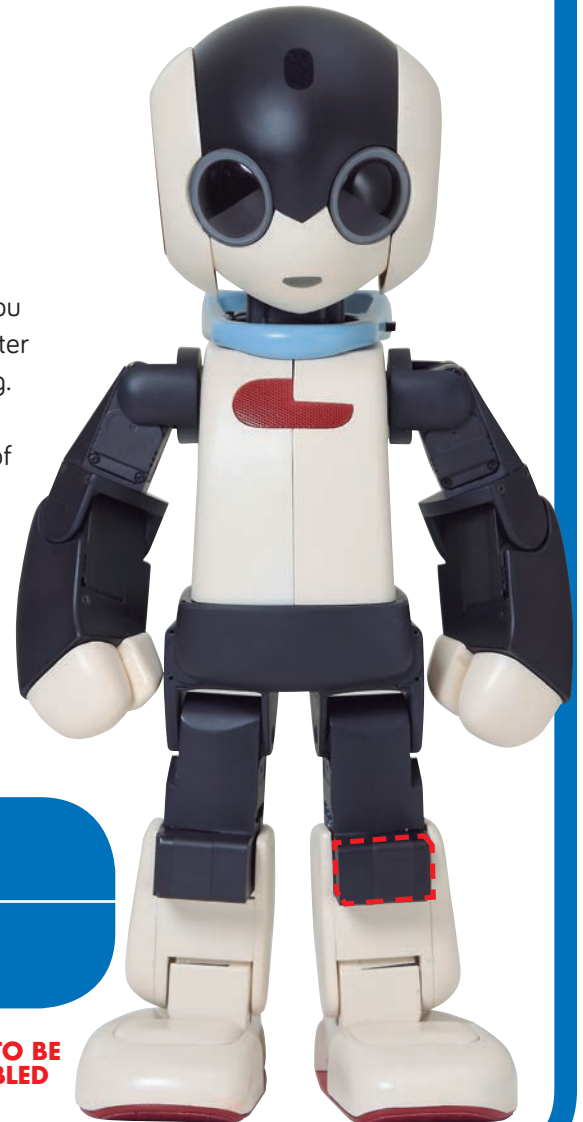


STAGE 47: PREPARE ROBI'S LEFT KNEE SERVO

In this assembly, you will prepare, test and set the servo motor that will operate Robi's left knee, servo ID=9.

The first assembly stage of this pack begins with the fairly straightforward task of wiring and programming another of Robi's 20 servos, servo ID=9, which works his left knee. The steps are the same as for many of his other servos, but again, be sure to

follow the instructions given here exactly, as if you make an error you may have to take the leg apart later on, which can be time-consuming. For this reason, it's important to check the ID number at the end of the stage (see Steps 13-14).



YOUR PARTS

1 Servo motor (ID=9)

**PARTS TO BE
ASSEMBLED**

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



*Servo cable
(Stage 46)*

*Neck stand/test board
(Stage 8)*



THE SERVO AND CABLE

TIP!

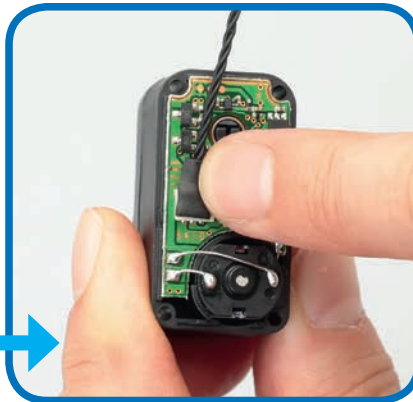


As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath.

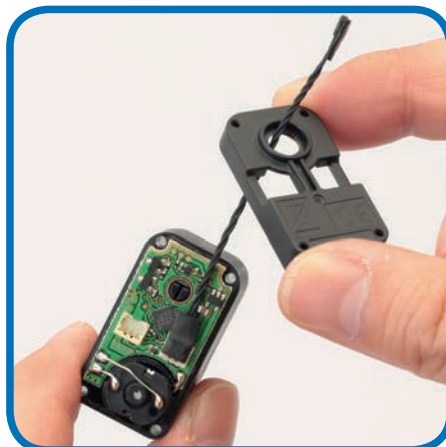


TIP!



This servo cable had protective pads fitted to its connectors in the previous stage, but if you find it difficult to attach with the pad in place, it is possible to apply the pad afterwards. If you choose to do the assembly this way, be especially careful not to press down on the circled projection, next to the cable, as you do so.

3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which end of the servo cable you use.

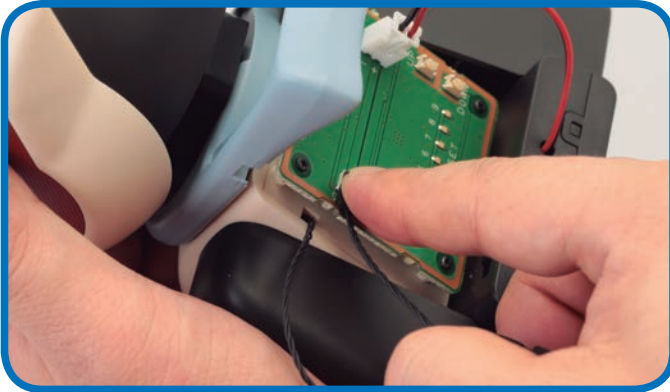


4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

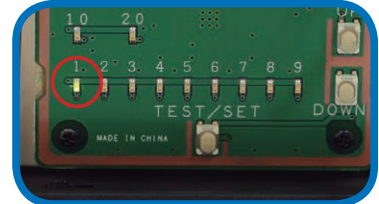
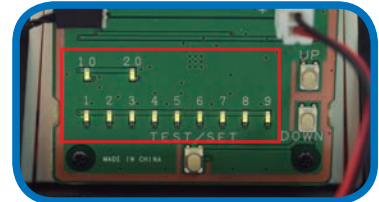
5 Press the cover back into place on the servo's main housing.

6 Fasten the cover back in place with all four screws, but only loosely as you will be removing them again at a later stage.

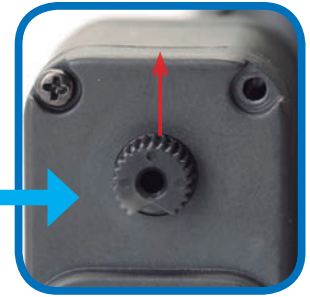
TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board, and attach the servo cable connected to the servo in Step 3.

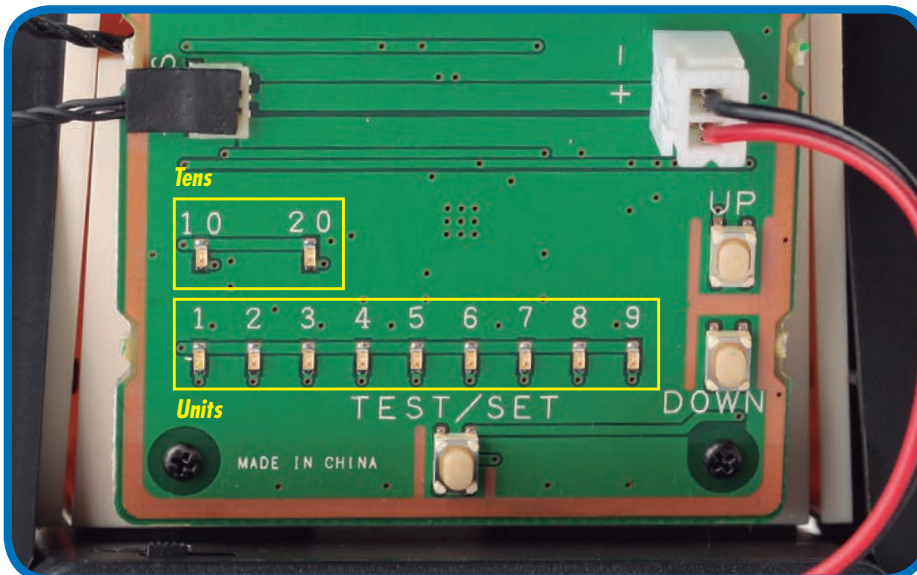


- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

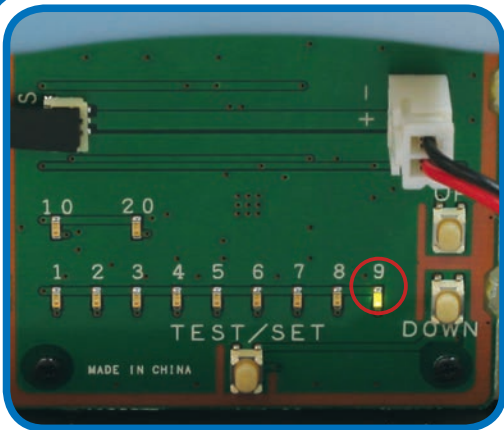


- 9** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

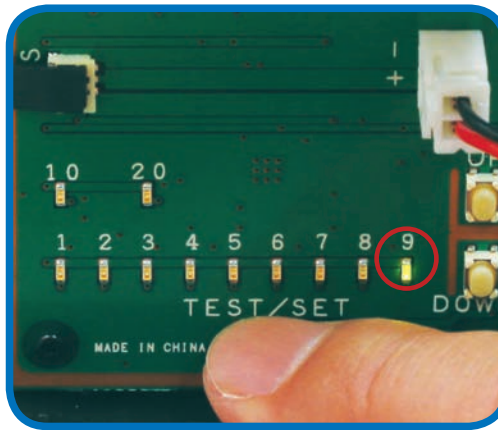
SETTING THE ID NUMBER



- 10** Re-familiarise yourself with the lights and switches on the test board. These are used to set the ID numbers of the servos. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 9, as you will do in this stage, you press UP 8 times. This will light up the 9 LED alone. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.



11 To set the ID number of Robi's left knee servo to 9, press the UP button 8 times. Make sure that the 9 LED alone is lit, using the UP and DOWN buttons to correct the number if necessary.

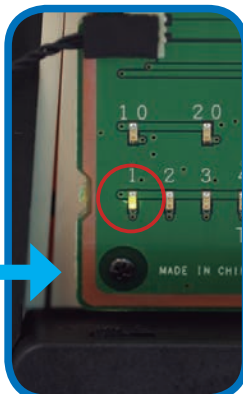
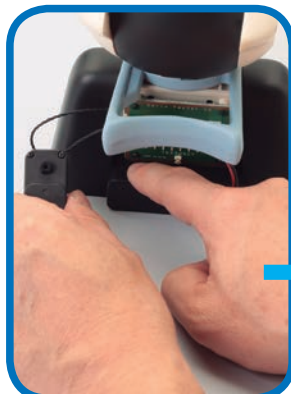


12 Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 9 light should flash a few times, then remain lit.

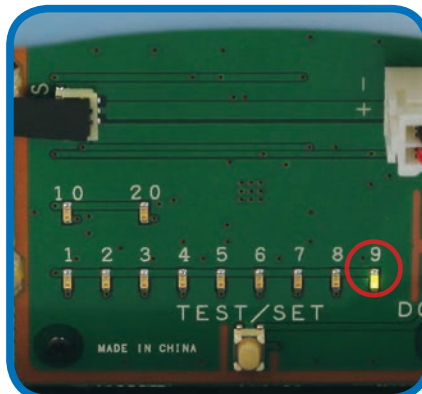


ID=9

CHECKING THE ID NUMBER



13 It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.

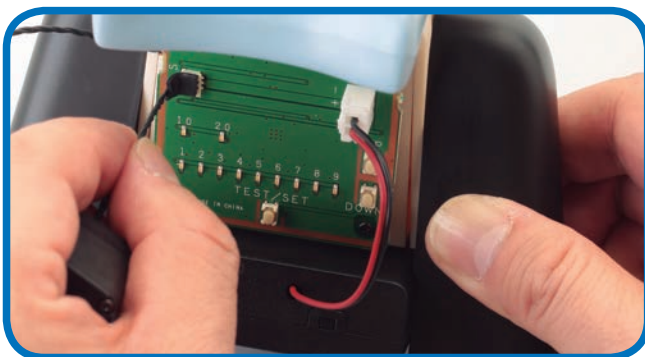


14 To check the ID number, press TEST/SET. Now the 9 LED should light up alone. Only press TEST/SET briefly – do not hold it in, or the servo's ID number may be wiped. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

HELP!



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake or set the wrong ID number, you can reset it by going through the setting procedure again.



15 Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit to Robi's leg.

Assembled knee servo

The servo that will operate Robi's left knee is assembled, wired and set. Make sure you keep this part safely, preferably in a sealed plastic bag with its ID number (9) written on it for ease of reference.



STAGE 48: ASSEMBLE ROBI'S LEFT KNEE, THEN TEST HIS LEFT LEG

In this session, you will fit the left knee servo prepared in the previous stage into the knee frame supplied with this stage. You will then join the knee subassembly to the rest of the left leg, and test it using the CPU board.

This stage sees you again linking pre-wired servos and handling open servo motors. As ever, great care must be taken when working with the delicate electronic components and

cables, as they are easily damaged. Once you have joined the knee servo to the rest of the left leg, you will proceed to test its movement as you have been doing over the previous stages.



PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Left knee frame
- 2 Left inner lower leg panel
- 3 M2 x 7mm countersunk screws x 3 (one is a spare)
- 4 Servo cable 70mm

SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.



*Servo ID=9
(Stage 47)*

*Battery
(Stage 38)*



*CPU board, with test micro
SD card still inserted
(Stage 44)*

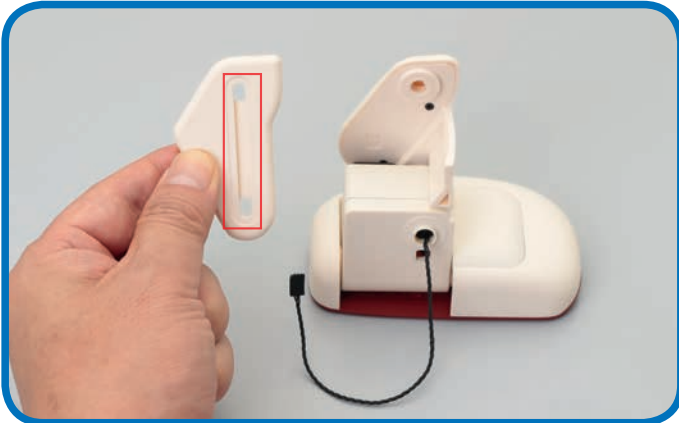


*Left foot and ankle
assembly (Stage 46)*

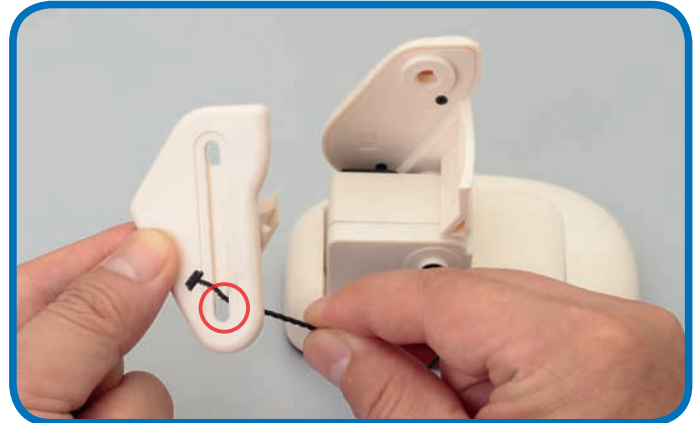


*Protective pads
(Stage 3)*

BUILDING UP THE LOWER LEG



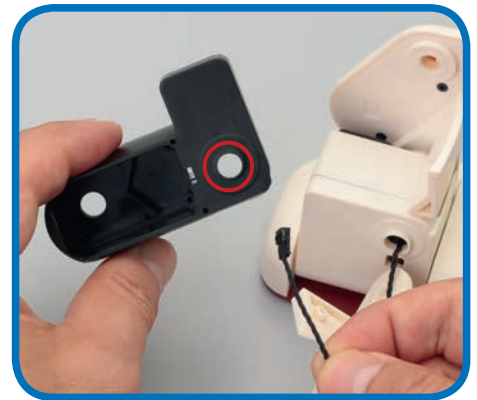
- 1 Set Robi's left foot and ankle assembly out on your work surface, with the ankle perpendicular to the foot's sole. Line up the inner lower leg panel with the outlined groove facing you, as shown.



- 2 Feed the servo cable running out from the hole in the ankle frame into the lower hole in the leg panel, from the inside of the part.



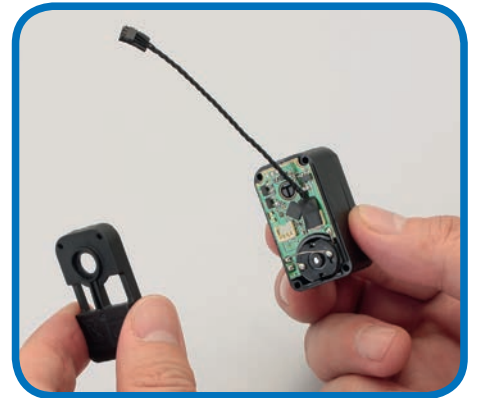
- 3 Now bend the servo cable over and pass its free connector back to the rear of the inner leg panel, by feeding it through the hole at the top of the part.



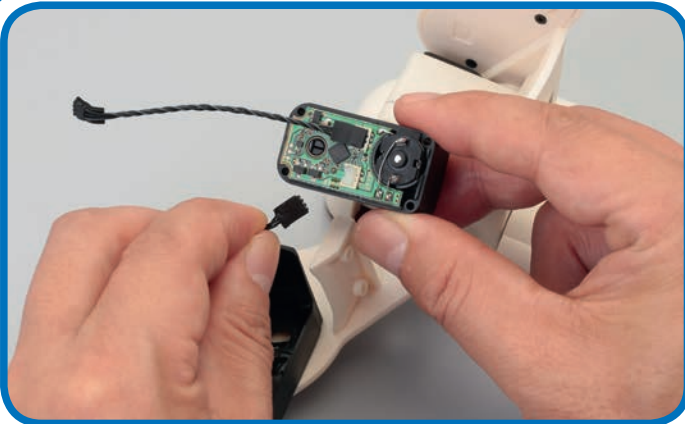
- 4 Now line up the knee frame, orientated as shown above, to the servo cable's connector. The circled hole on the inside of the frame should face you.



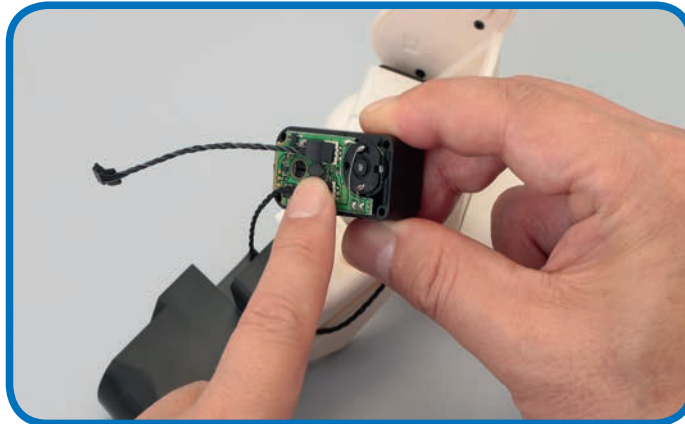
- 5 Feed the cable into the hole and pull it into the inside of the knee frame. Set the assembly down carefully on your work surface, making sure the cable remains in position.



- 6 Take servo ID=9, prepared in the previous stage, and remove its back cover. You will not be using the cover again, but keep the screws.



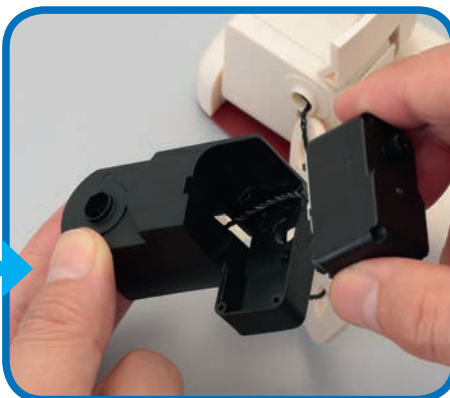
- 7** Position the open knee servo next to the left foot and ankle subassembly, making sure to hold the servo by its sides and taking care not to disturb any of the exposed electronics or the cable's connector.



- 8** Fit the connector of the ankle servo's cable into the free socket on the knee servo. Again, be careful not to interfere with the servo's components, or dislodge the first connector. As ever, be very careful when handling the servo cable connector: it will click into the socket neatly when set correctly – never force it!



- 9** Making sure both connectors remain securely in their sockets, feed the free end of the 70mm servo cable running from servo ID=9 into the rectangular hole in the knee frame.



- 10** Seat the servo into the knee frame. Gently pull on the both cables as you do this to minimise slack inside of the knee frame.

FITTING THE SERVO



- 11** Once the slack in the cables has been minimised, press the servo into the knee frame so that no gaps appear around its edges.



- 12** Use the four screws removed in Step 6 to secure the servo to the frame.

FITTING THE KNEE FRAME



13 Line up the shaft of servo ID=9, now set into its frame, to the inside face of the outer lower leg frame. The shaft will fit into the D-shaped hole in the frame (circled), with the body of the servo sticking out above Robi's 'shin'. As this is another D-cut hole, it is important that you align the parts correctly from the start.

Making absolutely sure the servo's shaft is sitting neatly into the D-cut hole (see Close-Up box, below), press the parts together to join them.

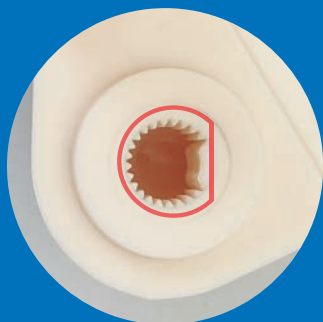
14



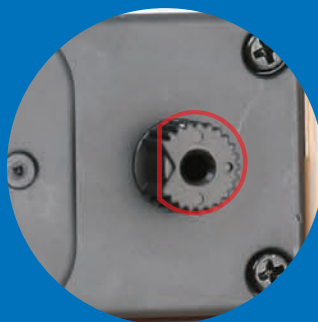
THE D-CUT



By now you will know how important it is that you take care when working with the servos and frames. Each joint, such as the one linking the lower leg and knee frame fitted in this stage, has a precisely moulded D-cut hole, designed to fit snugly onto the correspondingly shaped servo shaft to ensure that the servo is set correctly from the start. When these parts are aligned as they should be, it will be easy to fit them together. If it is in any way difficult to join the parts, separate them, check that you have both of them lined up accurately, and try again. Never force the parts together, as this could damage both of them.



The D-cut hole in the lower leg frame, supplied with this stage.



The servo's shaft is also D-shaped, and is designed to fit snugly into the hole in the frame.

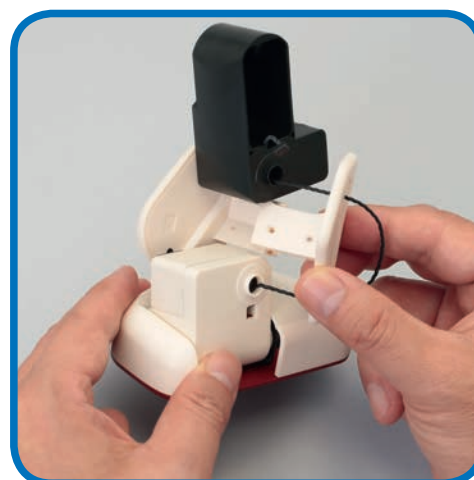
Never force the parts together, because there is a high risk that you will damage the small plastic teeth inside the hole, as seen below.



FITTING THE LOWER INNER LEG PANEL

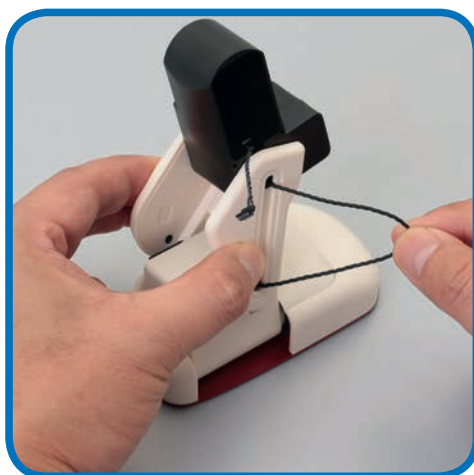


- 15** Turn the inner lower leg panel to face you – there are two circular indents around each hole. These will accommodate the circular ridges on the knee and ankle frames (arrowed).



- Fit the panel into place, making sure the cable does not get pinched as you do so.

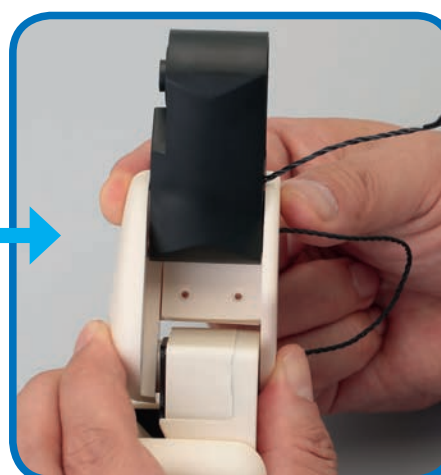
16



- 17** Gently pull the middle of the servo cable, as shown above, so that there is no slack inside the leg assembly.



- 18** Check from the top and back that no part of the servo cable has been pinched when you fitted the leg panel to the ankle and knee frames. The parts should be completely flush with one another.



- 19** Check that the two circled screw holes in the lower leg panels are aligned.



- 20** Insert and tighten two M2 x 7mm screws into these holes to secure the parts. Tilting the leg forward a little may help you do this.



TESTING THE LEFT LEG

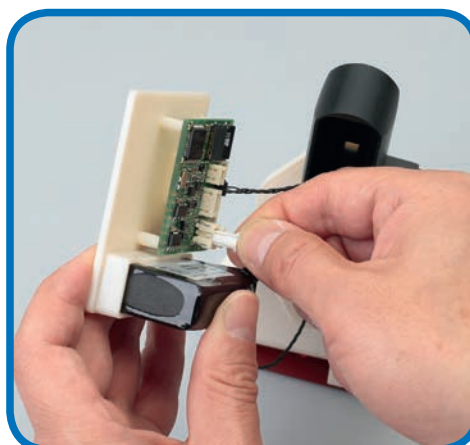
ATTENTION!

Robi's left leg will start moving as soon as you begin the test, so make sure you perform this test in a safe place where the leg will not knock anything over.

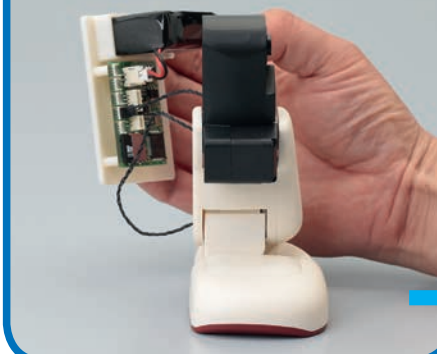
WARNING!

When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!

- 21** Insert the knee servo's free cable into the left leg socket on the CPU, which is the one second-closest to the micro SD card slot.

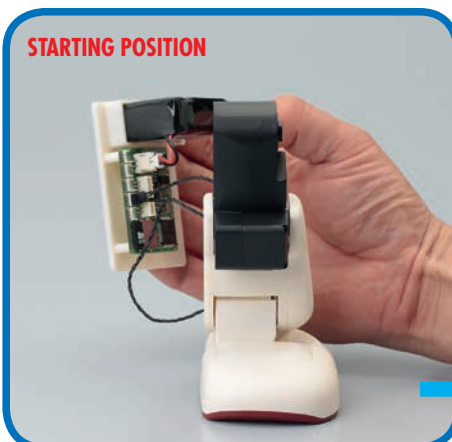


STARTING POSITION

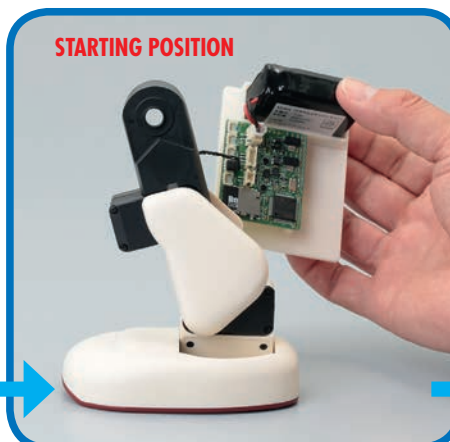


- 22** Set the leg flat on your work surface, then connect the battery – **the test will start as soon as the power is connected**, so be prepared and observe the following steps.

- 23** Watch from the front to see the leg move to its starting, or neutral, position, then begin the test by tilting inwards and back using the foot servo.



STARTING POSITION



STARTING POSITION



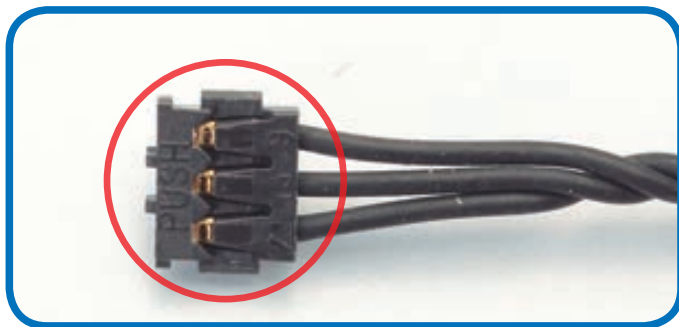
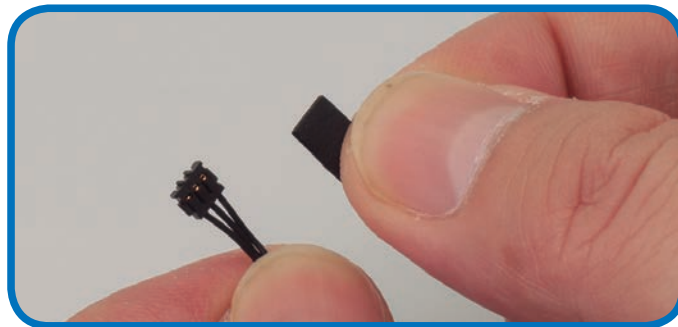
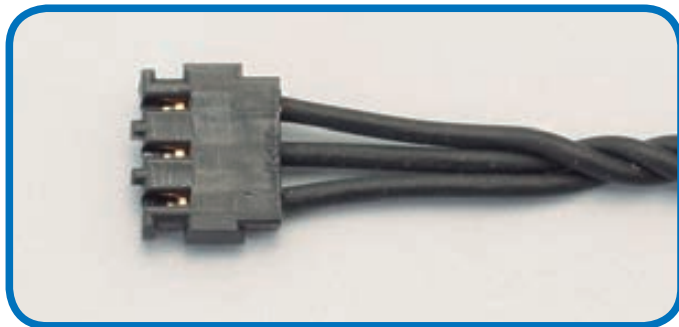
- 24** Now view the leg from the side to see it revert to its starting position, before beginning the test of the ankle and knee servos. In this part of the test the leg will tilt forward and back at the ankle and knee, before returning to the starting position. The test will repeat until you disconnect the battery.

Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

If the test does not run as described, read the troubleshooting section in this stage to see how to address the issue.

SEALING THE SERVO CABLE



25 As you have done previously, identify the side of this stage's servo cable connector that has PUSH marked on it (circled).

26 Apply a protective pad to the connector on this side. Repeat for the other end of the cable.

TIP!

If you want to wait to fit the pads until the cable is used in the build, you can do so.



Assembled left leg

At the completion of this stage, Robi's left leg is really beginning to take shape.

NOTE: On completion, the length of servo cable sticking out between Robi's ankle and knee will be pulled taut and sit neatly within the groove marked below by the dotted line. Until then, be very careful not to snag or hook this cable on anything, as this will damage the part and result in a great deal of dismantling and rebuilding!



NOTE: As with other stages in which you connected two or more servos, it is vital that you do not attach the servos linked in this stage to the test board! Doing so will wipe their ID numbers, meaning you will have to go back and open each servo to reset it.

TROUBLESHOOTING

As you may already have found with Robi's other limbs, there is a chance that the left leg will not work as intended the first time it is switched on. If the movement test carried out in this stage does not go as described, the most likely cause will be an error with one of the subassembly's servos or connections. Read this troubleshooting guide to see how to assess and correct any potential issues.

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:

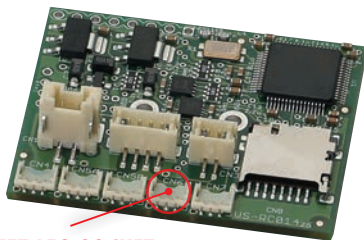


Neck stand (Stage 8)

IF NONE OF THE SERVOS MOVE

Before dismantling the leg to check the individual servos, check the other electrical components used in the test. Then, if that doesn't solve the problem, here's how to dismantle the subassembly safely and efficiently.

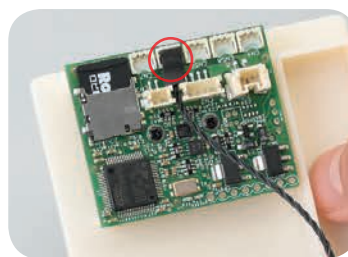
CHECK THAT THE LEG IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



LEFT LEG SOCKET

The socket second-closest to the micro SD card slot on the CPU board will hold the left leg's servo cable.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the left leg's servo cable connector is sitting securely within its socket. Remove and replace it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged by following the steps explained in Stage 41.

IF NO SERVOS MOVE

If none of the servos move when you perform the test, it might just be a connection fault. If not, you may need to dismantle and check all the servos' connections and, if necessary, ID numbers.



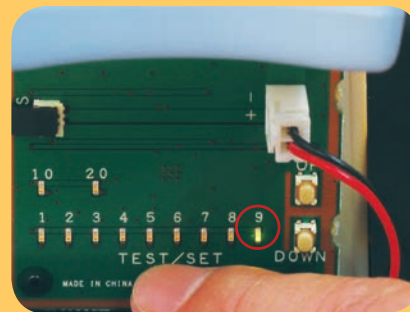
DISASSEMBLY

Follow the assembly of Steps 7-20 of this stage in reverse, and check the connections carefully, starting with the knee servo. If this does not solve the issue, you will need to continue to dismantle the ankle and foot and test/reset their

faulty servo(s) as necessary. See the 'Testing the ankle and foot servos' section of Stage 45 for more on this.

IF THE KNEE SERVO DOES NOT MOVE

If the knee servo does not move, but the other two do, it is only the knee servo that needs attention. Follow Steps 7-20 of this stage in reverse and check the knee servo's ID number, resetting it if necessary.



DISASSEMBLY

Follow the assembly of Steps 7-20 of this stage in reverse, and check the connections carefully. If this does not solve the issue, you will need to test/reset the knee servo. Its ID number

should be 9, and more information on the process of resetting this servo's ID number is found in its original stage, Stage 47.

STAGE 49: BUILD UP AND TEST ROBI'S LEFT LEG

Set the ID number of Robi's left thigh servo, then fit and test it as part of the left leg assembly built over the previous stages.

In this session, you will program the servo that will move Robi's left upper leg back and forth, servo ID=8. The assembly is the same as for the right leg in Stage 31, but this time, you will also test the servo once fitted, using

the CPU board. The process of setting the servo's ID number is the same as for all of the other servos, but, as ever, be sure to follow the steps exactly, as going back to repair any errors can be time consuming.



1



YOUR PARTS

1 Servo motor (ID=8)

**PARTS TO BE
ASSEMBLED**

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Servo cable
(Stage 48)



Battery (Stage 38)



CPU board with test
micro SD card inserted
(Stage 44)



Left leg assembly
(Stage 48)



Neck stand
(Stage 8)

THE SERVO AND CABLE

TIP!

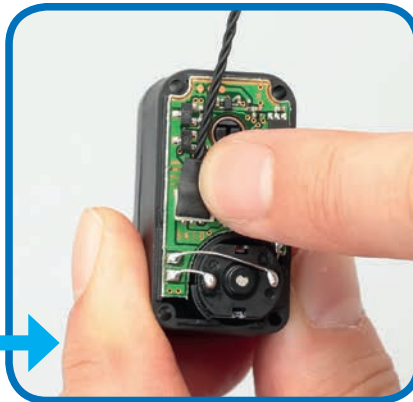


As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

2 Gently lift the cover away, taking care not to force it or to damage the circuit board underneath.



TIP!



This servo cable had protective pads fitted to its connectors in the previous stage, but if you find it difficult to attach with the pad in place, it is possible to apply the pad afterwards. If you choose to do the assembly this way, be especially careful not to press down on the circled projection, next to the cable, as you do so.

3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which end of the servo cable you use.

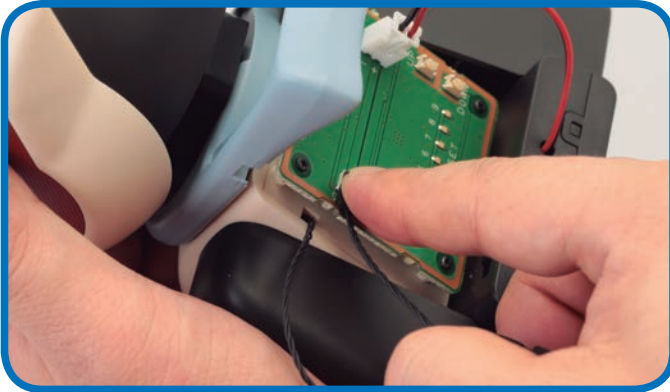


4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

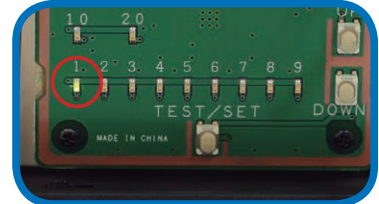
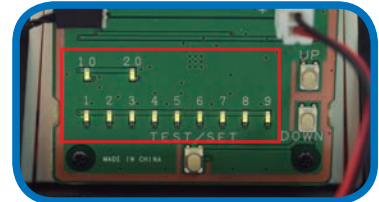
5 Press the cover back into place on the servo's main housing.

6 Fasten the cover back in place with all four screws, but only loosely as you will be removing them again at a later stage.

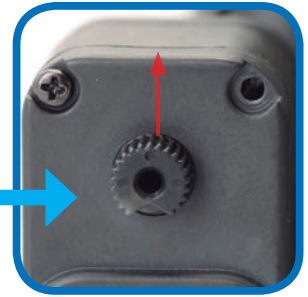
TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board and attach the servo cable connected to the servo in Step 3.

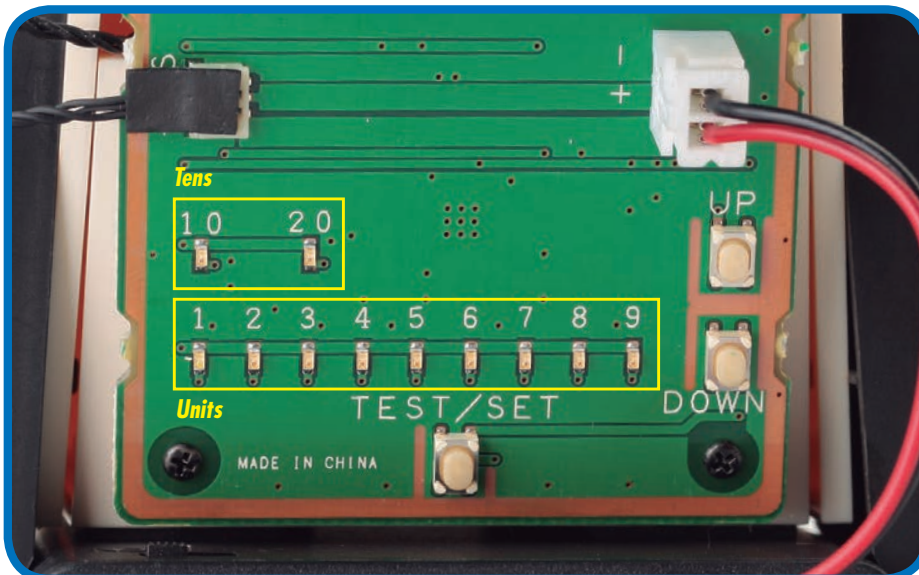


- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

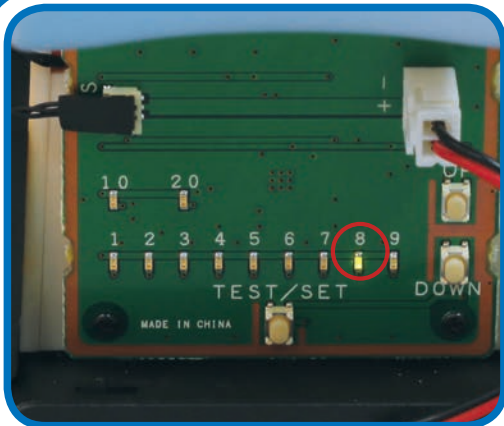


- 9** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

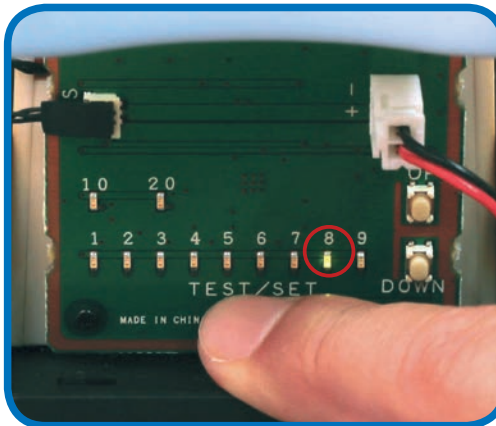
SETTING THE ID NUMBER



- 10** Next you will set the ID number of the servo using the test board, as you have done before. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 8, as you will do in this stage, you press UP 7 times. This will light up the 8 LED alone. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.



11 To set the ID number of Robi's left thigh servo to 8, press the UP button 7 times. Make sure that only the 8 LED is lit, using the UP and DOWN buttons to correct the number if necessary.

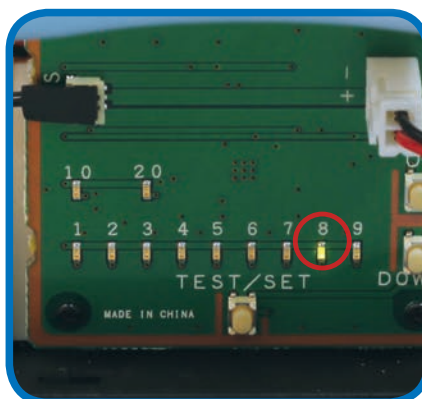
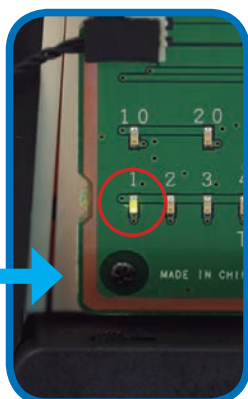
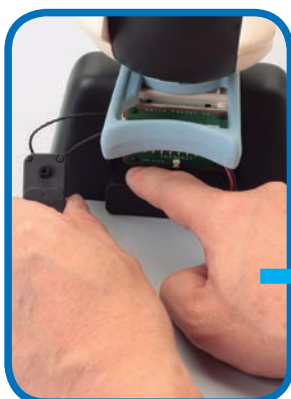


12 Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 8 LED should flash a few times, then remain lit.



ID=8

CHECKING THE ID NUMBER



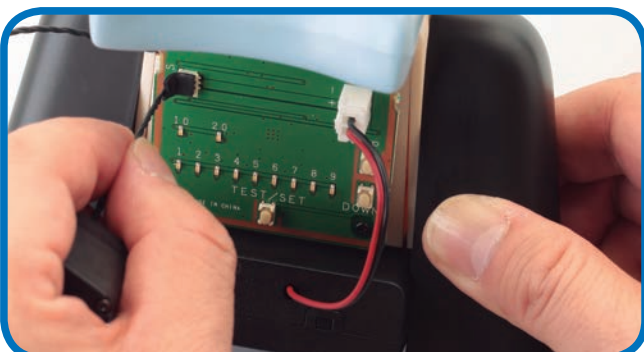
HELP!



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake, or set the wrong ID number, you can reset it by going through the setting procedure again.

13 It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.

14 To check the ID number, press TEST/SET briefly. Now the 8 LED should light up and the servo shaft should repeat the rotations seen in the test. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.



15 Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit into Robi's leg.



16 Remove the screws fitted in Step 6 and take the cover off. Keep the screws, but you will not be using the cover again.

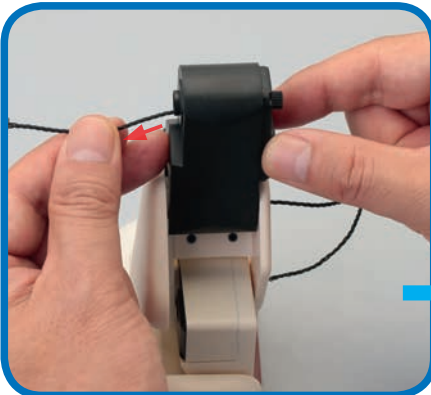
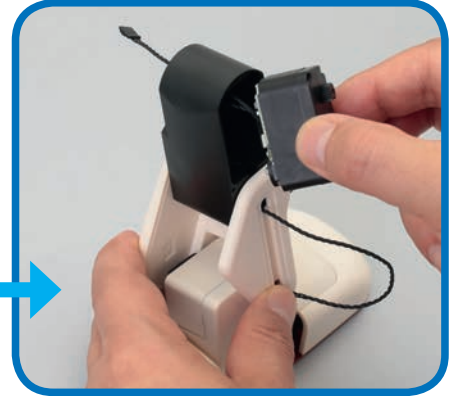
FITTING THE THIGH SERVO



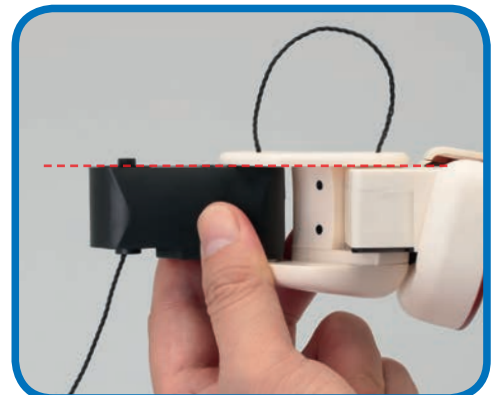
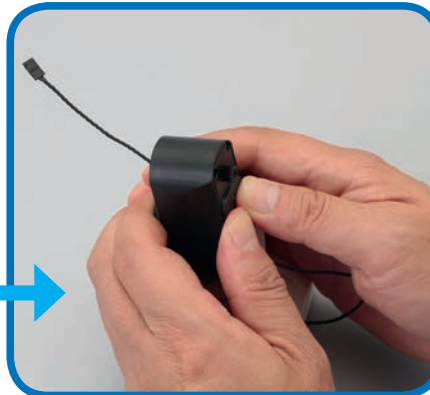
- 17** Line up the thigh servo prepared in the previous steps to the left leg assembly. Then, very carefully, press the connector of the cable running out from the knee servo to the free socket on the thigh servo, as shown.



- 18** Now feed the free tip of the thigh servo's cable through the circled hole in the knee frame, from the inside of the part. Then move the servo into the correspondingly shaped frame, making absolutely sure that you do not trap the servo cables as you do so.



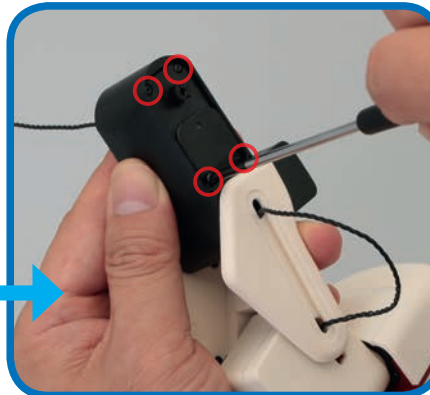
- 19** Press the servo into the frame, so that it sits flush with the edges of the part. Make sure that the servo cable leading out of the thigh servo does not get caught or trapped, by pulling it gently through from the outside as you do this, and perform this step very gently to ensure both connectors remain in place.



- 20** View your assembly from the side to double-check that the parts are sitting flush with each other.



- 21** Use the four screws that originally held the thigh servo cover in place to secure the parts, as shown.



NOTE: As with other stages in which you connected two or more servos, it is vital that you do not attach the servos linked in this stage to the test board! Doing so will wipe their ID numbers, meaning you will have to go back and open each servo to reset it.

Assembled left leg

The servo that will operate Robi's left thigh movement is now in place, ready to be joined to his hip in the following stages.



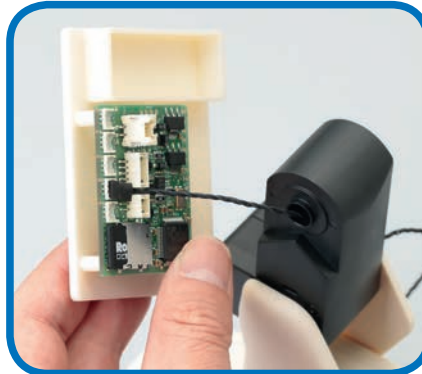
TESTING THE LEFT LEG

ATTENTION!

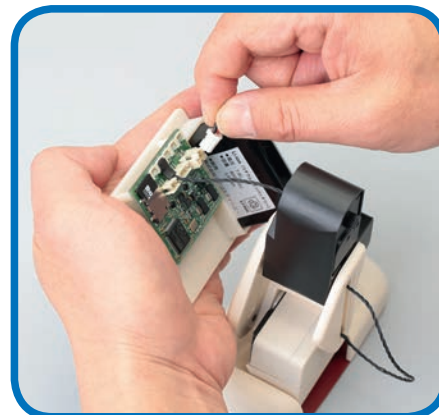
Robi's left leg will start moving as soon as you begin the test, so make sure you perform this test in a safe place where the leg will not knock anything over.

WARNING!

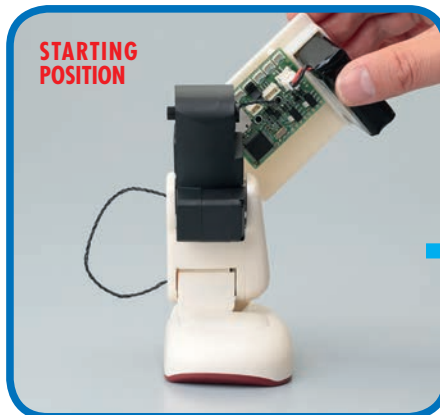
When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!



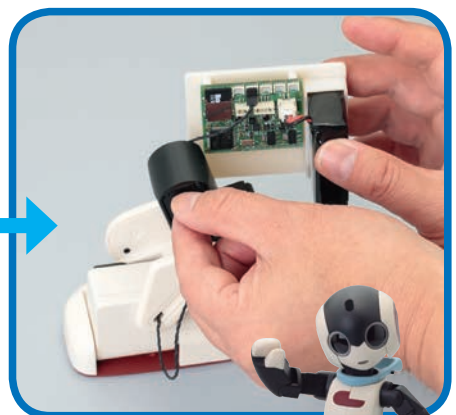
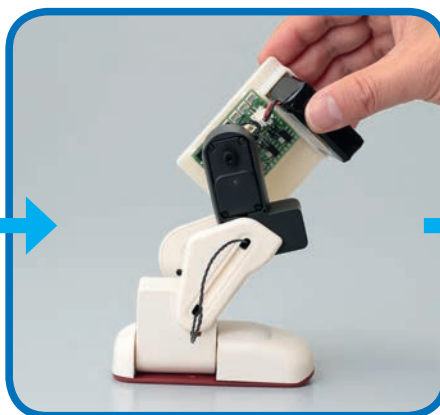
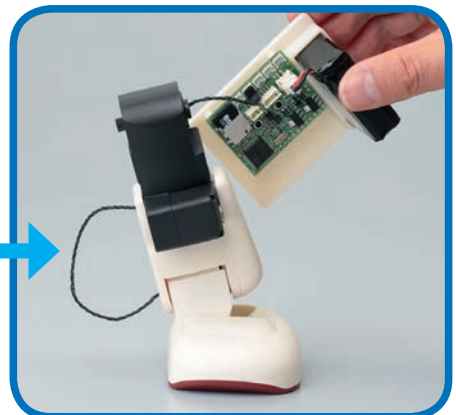
- 22** Insert the servo cable leading from the left leg into the left leg socket on the CPU, second-closest to the micro SD card slot.



- 23** Set the left leg flat on your work surface, then connect the battery –remember, the test will start as soon as the power is connected.



- 24** Watch from the front to see the leg move to its starting, or neutral, position, then begin the test by tilting inwards and back using the foot servo.



- 25** The leg should then revert to its starting position, before beginning the test of the ankle, knee and thigh servos. This part of the test is best observed from the side, and will include the leg tilting forward and back at the ankle, knee and thigh servos. As no further components are fitted to the thigh servo at this point, touching the shaft as this happens will be the best way to affirm its movement, as seen above far right. The test will repeat until you disconnect the battery.

Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

If the test does not run as described, read the troubleshooting section in this stage to see how to address the issue.

TROUBLESHOOTING

In this section, we look again at what to do if the movement test carried out in this stage does not go as described.

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:



Servo removal tool (Stage 41)

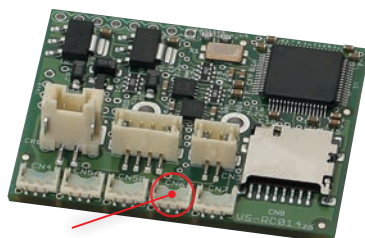


Neck stand (Stage 8)

IF NONE OF THE SERVOS MOVE

Before dismantling the leg to check the individual servos, check the other electrical components used in the test. Then, if that doesn't solve the problem, here's how to dismantle the subassembly safely and efficiently.

CHECK THAT THE LEG IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



LEFT LEG SOCKET

The socket second-closest to the micro SD card slot on the CPU board will hold the left leg's servo cable.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the left leg's servo cable connector is sitting securely within its socket. Remove and replace it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged by following the steps explained in Stage 41.

IF NO SERVOS MOVE

If none of the servos move when you perform the test, it is likely that there is a connection fault, and resetting this may resolve the issue. If not, you may need to dismantle the leg and check all the servos' connections and ID numbers.



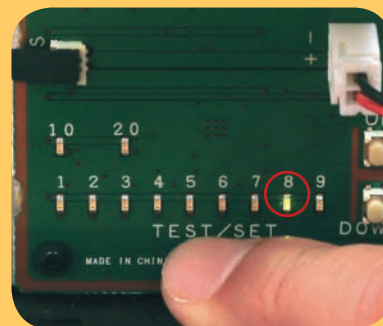
DISASSEMBLY

Follow the assembly of Steps 17-21 of this stage in reverse, and check the connections carefully, starting with the thigh servo. If this does not solve the issue, you may need to continue to dismantle

the knee, ankle and foot and test/reset their faulty servo(s) as necessary. See the relevant sections of Stages 45 and 48 for more on this.

IF THE THIGH SERVO DOES NOT MOVE

You tested the left foot, ankle and knee servos in the previous stage, so these should be sound (see Stage 48 if this is not the case). If, however, the thigh servo does not move, follow Steps 17-21 of this stage in reverse and check the thigh servo's connectors and, if necessary, test/reset its ID number.



DISASSEMBLY

Follow the assembly of Steps 17-21 of this stage in reverse, and check the connections carefully. If this does not solve the issue, you will need to test/reset the thigh servo. Its ID number should be 8, and more information

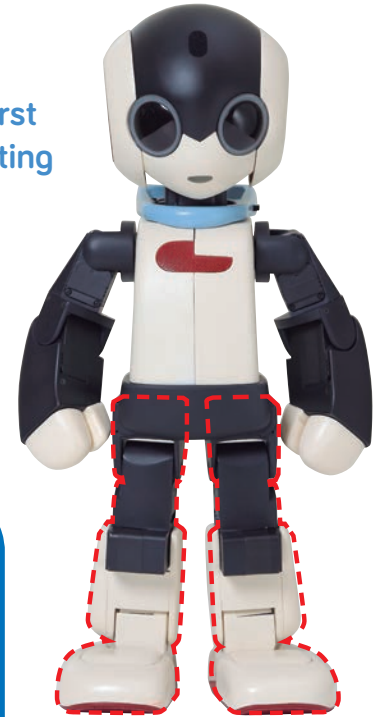
on the process of resetting this servo's ID number is found earlier in this stage.

STAGE 50: PREPARE TO JOIN ROBI'S LEGS FOR THE FIRST TIME

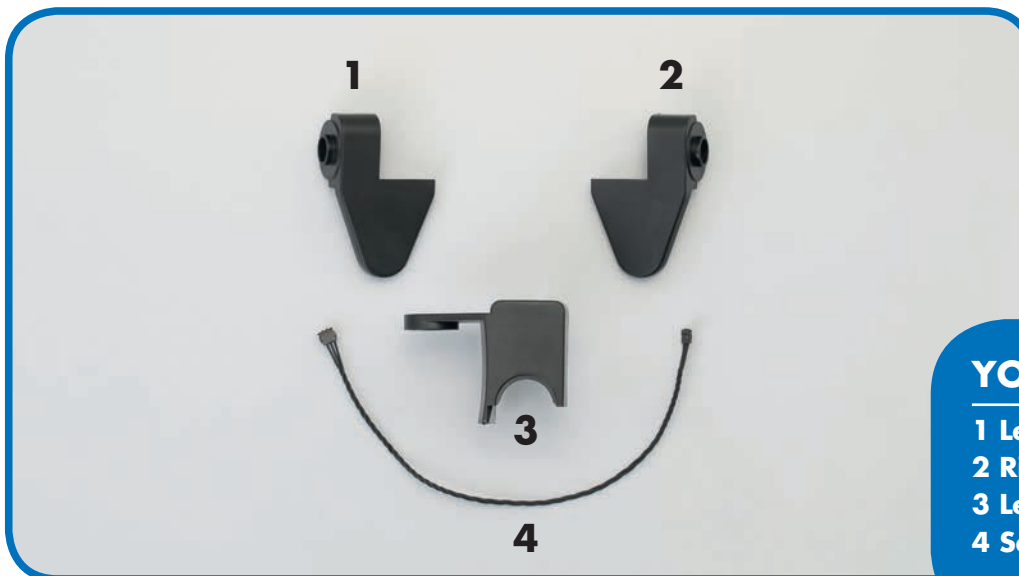
In this stage, you will begin the process of joining Robi's legs for the first time, which you will do by fitting his left leg with its hip frame, then fitting the right hip servo programmed in Stage 33 to the existing hip frame of the right leg.

This stage begins with you adding the inner hip frame to the top of the left leg assembled over the previous stages. You will then turn to the right leg, where you will fit the right hip servo prepared in

Stage 33 (ID=2) to the existing hip frame at the top of the right leg. This will then be tested using the CPU board and rear hip panel, following the same method used in previous stages.



PARTS TO BE ASSEMBLED

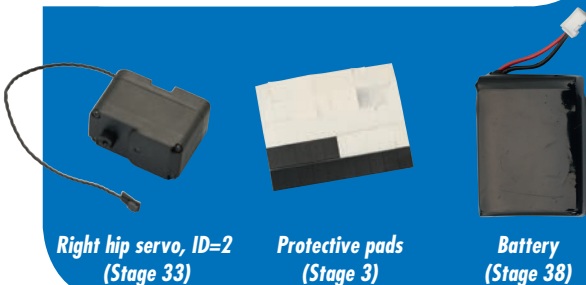


YOUR PARTS

- 1 Left outer hip frame
- 2 Right outer hip frame
- 3 Left inner hip frame
- 4 Servo cable 135mm

SAVED PARTS

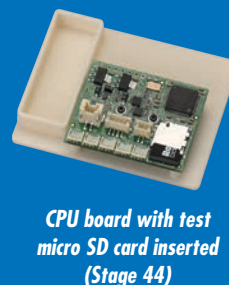
You will be using parts from previous stages, so be sure to have these to hand before beginning.



Right hip servo, ID=2
(Stage 33)

Protective pads
(Stage 3)

Battery
(Stage 38)



CPU board with test
micro SD card inserted
(Stage 44)



Right leg
(Stage 32)

Left leg
(Stage 49)

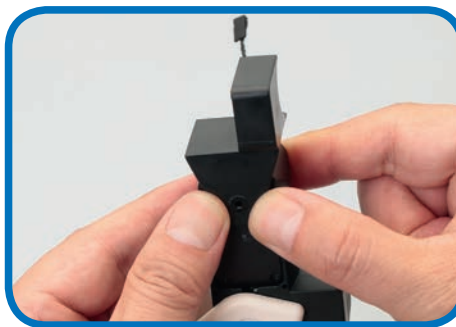


Rear hip panel
(Stage 40)

MOUNTING THE INNER HIP FRAME



- 1 Line up the left inner hip frame to the left leg assembly built over the previous stages, oriented as shown. The D-cut hole in the frame will fit over the correspondingly shaped shaft of servo ID=8, the thigh servo.



- 2 Making absolutely sure the shaft is sitting correctly within the D-cut hole, press the parts together.



- 3 Double-check that the shaft and D-cut hole of the frame are correctly set and flush with one another.

TIP!



The D-cut is essential for setting the neutral position of Robi's moving parts. Never force the parts, as this can result in permanent damage. See the Close-Up box in Stage 48 for more details.

Assembled left leg



RETESTING THE SERVOS



You tested the movement of the left leg's servos in Stage 49, but at that stage, the thigh servo wasn't attached to the inner hip frame fitted in this stage. To get an idea of how the parts will work together now that they are joined, you may like to perform the test again. If anything is amiss and you need to dismantle the subassembly, refer to the relevant steps of Stage 49; if you find you need to reset the servos' ID numbers, refer to the relevant sections of Stages 45, 48 and 49. Remember, a diagram of each of Robi's servo ID numbers is found in Packs 3 and 11.



Connect the ankle servo's cable to its designated socket on the CPU board (see diagram in Stage 43). Then connect the battery.



The test will start immediately, beginning by returning the servos to their neutral positions, then tilting the foot servo from side to side.



Now you will see the thigh servo in action more clearly, as it will bend the inner thigh frame forwards then back, completing the test.

FITTING THE RIGHT HIP SERVO

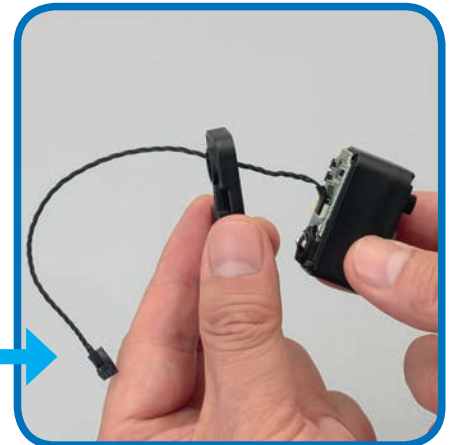
NOTE: you have been working on the left leg in this stage, but now you will start on the right leg.



4 Fetch the right hip servo (ID=2) that you prepared in Stage 33.



5 Remove the four screws holding the servo cover in place.



6 Position the right hip servo next to the right leg assembly. Hold the connector at the end of the cable running into the leg, and push it into the free socket on the servo.



7 Hold the servo as shown and line it up to the right leg's inner hip frame.



8 Press the servo into the frame. There should be no gap between the servo and the frame.



9 Use two of the screws removed in Step 5 to secure the servo to the hip frame via the two circled holes.



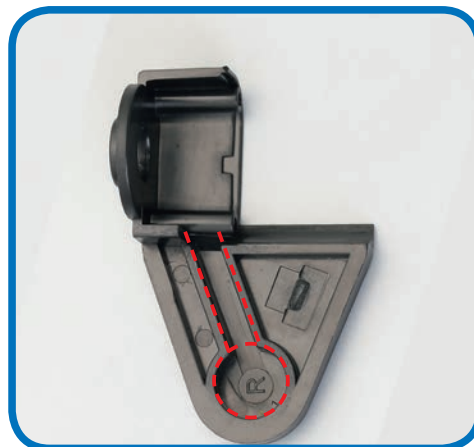
FITTING THE RIGHT OUTER HIP FRAME



- 10** Lay the two outer hip frames supplied in this stage side by side on your work surface. You will see an 'L' and an 'R' moulded into the inside faces of the left and right parts respectively (circled). In this stage, you will be using only the **right** frame, so store the left one safely until it is needed.



- 11** With the right outer hip frame oriented as shown, feed the free end of the hip servo's cable through the hole in its centre, from the inside of the part.



- 12** Once fitted, the cable will sit in the recess outlined above, so be aware of this before you join the parts.

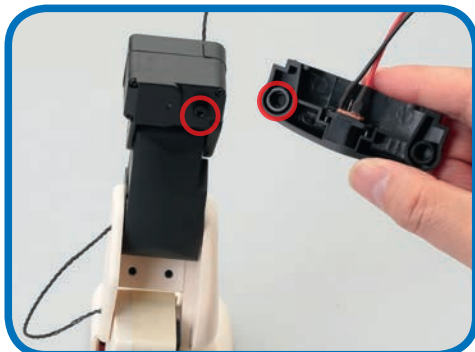


- 13** The next step is to join the outer hip frame to the rest of the hip assembly. Position the part close to the assembly, and seat the servo cable linking the thigh and hip servos into the groove highlighted in the previous step. Then, ensuring that the cable remains set this way, carefully join the parts so that they fit snugly. The lip of the thigh frame will sit inside the circular recess also outlined in the previous step.

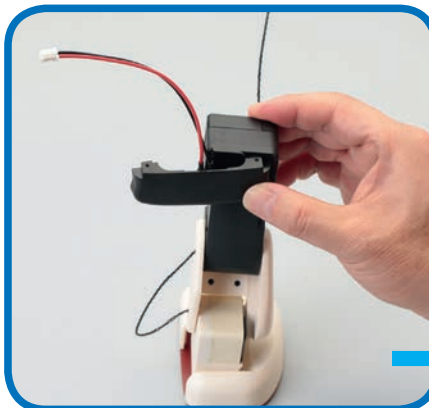


- 14** Use the remaining two screws removed in Step 5 to secure the frame to the servo.

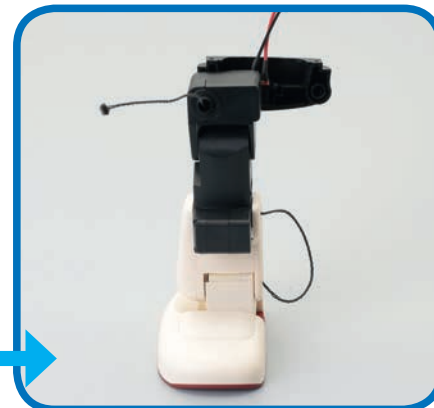
TESTING THE RIGHT LEG AND HIP



- 15** Fetch the rear hip panel (with the power distribution board and cord) and line up its circled D-cut socket to the backward-facing hip servo's shaft. Remember, the D-cut parts will fit neatly and easily when set correctly.

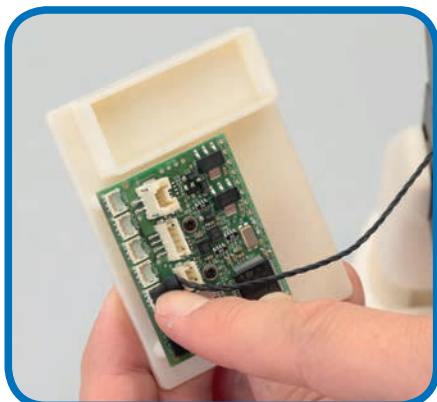


- 16** When the parts are set as they should be (see the Close-Up boxes throughout the series for info on handling D-cuts safely), press the hip panel onto the shaft, then turn the leg around to face you.

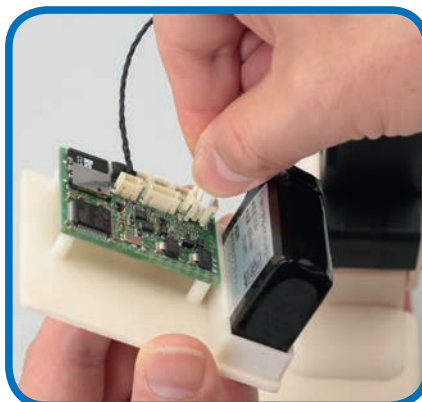


ATTENTION!

Robi's right leg will start moving as soon as you begin the test, so make sure you perform this test in a safe place where the leg will not knock anything over.



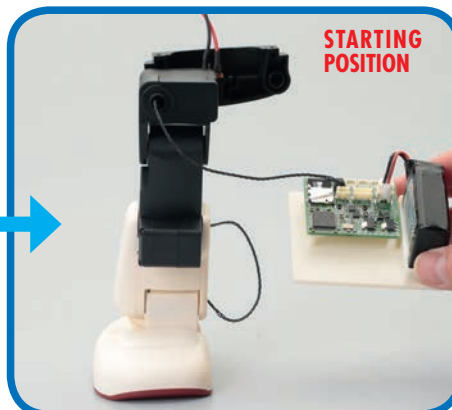
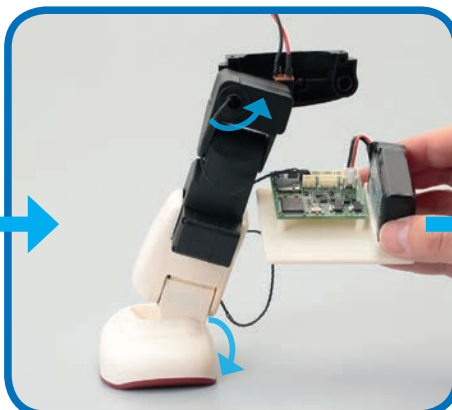
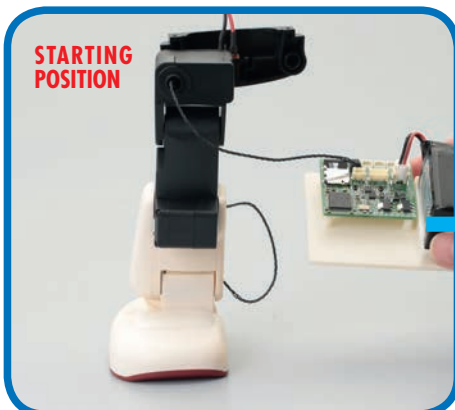
- 17** Now fetch the CPU board with the test micro SD card inserted, and insert the hip servo's cable into the right leg socket on the board, closest to the micro SD card slot.



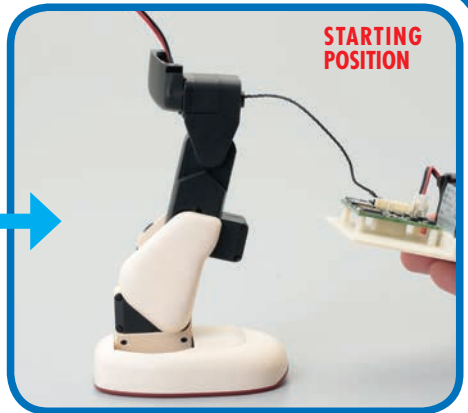
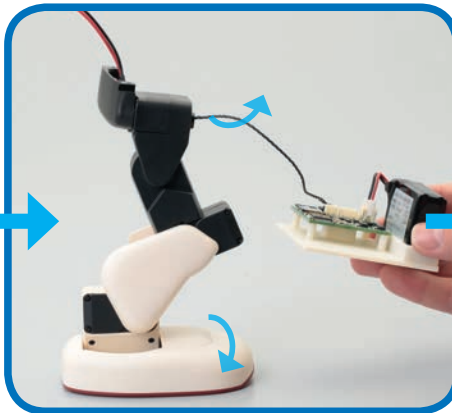
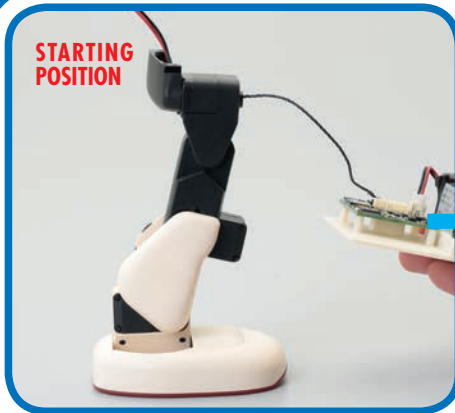
- 18** Connect the battery to initiate the test – bear in mind that **the test will begin as soon as the battery is connected.**

WARNING!

When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!



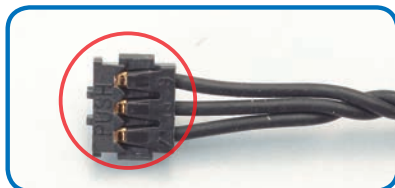
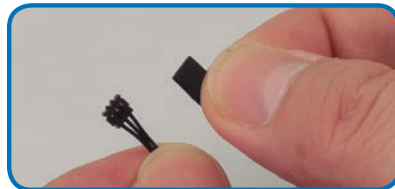
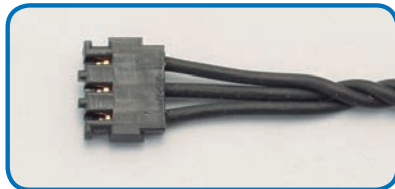
- 19** When the test begins, the leg will move to its starting, or neutral, position. Then, the foot and hip servos – as the two that rotate from side to side – will move together, with the hip servo fitted in this stage counteracting the inward tilt of the foot servo to keep the hip panel level. The servos will then return to their starting positions once more.



- 20** Next, the three servos that tilt forward and back – the ankle, knee and thigh servos – will begin their movement test by adjusting themselves to make Robi's right leg 'squat' down for a moment. They will then return to their starting positions. The test will continue to run until the battery is disconnected, so observe a few runs of it, then remove the battery to end the test. When you are happy that all is as it should be, remove the rear hip panel and store it safely until it is next needed.

If the test does not run as described, read the troubleshooting section in this stage to see how to address the issue.

SEALING THE SERVO CABLE



- 21** As you have done previously, identify the side of this stage's servo cable connector that has PUSH marked on it (circled).

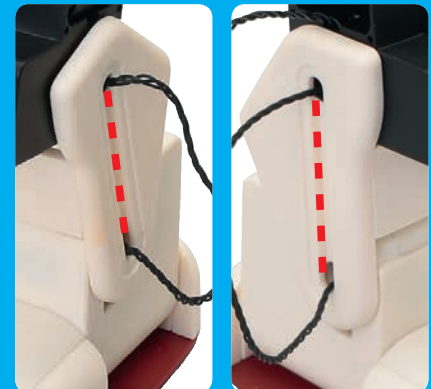
- 22** Apply a protective pad to the connector on this side. Repeat for the other end of the cable.

TIP!

If you want to wait to fit the pads until the cable is used in the build, you can do so.

Also, as with other stages in which you connected two or more servos, it is vital that you do not attach the servos linked in this stage to the neck stand/test board! Doing so will wipe their ID numbers, meaning you will have to go back and open each servo to reset it.

NOTE: On completion, the length of servo cable sticking out between Robi's ankles and knees will be pulled taut and sit neatly within the grooves marked below by the dotted line. Until then, be very careful not to snag or hook these cables on anything, as this will damage the parts and result in a great deal of dismantling and rebuilding!



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

Assembled right and left legs

Robi's legs have taken the next step towards being joined at the hip.



TROUBLESHOOTING

In this section, we look again at what to do if the movement test carried out in this stage does not go as described.

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:



Servo removal tool (Stage 41)

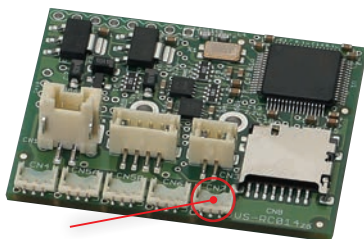


Neck stand (Stage 8)

IF NONE OF THE SERVOS MOVE

Before dismantling the leg to check the individual servos, check the other electrical components used in the test. Then, if that doesn't solve the problem, here's how to dismantle the subassembly safely and efficiently.

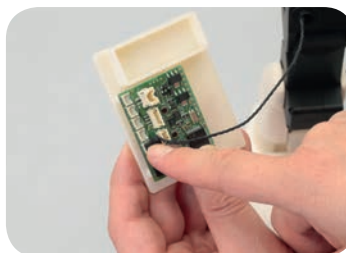
CHECK THAT THE LEG IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



RIGHT LEG SOCKET

The socket closest to the micro SD card slot on the CPU board will hold the right leg's servo cable.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the right leg's servo cable connector is sitting securely within its socket. Remove and replace it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged by following the steps explained in Stage 41.

IF NO SERVOS MOVE

If none of the servos move when you perform the test, it is likely that there is a connection fault with the hip servo, and checking/resetting its servo cable connections may well resolve this. If not, you will need to dismantle the leg and check the other connections and servo numbers.



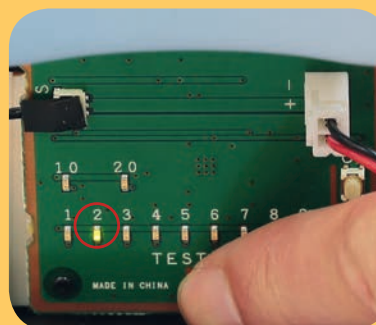
DISASSEMBLY

Follow the assembly of Steps 6-14 of this stage in reverse, and check the hip servo's connections carefully. If this does not solve the issue, you may need to continue to dismantle the thigh, knee, ankle and foot and test/reset

their faulty servo(s) as necessary. See the troubleshooting guide in Stage 46 for more on this.

IF THE HIP SERVO DOES NOT MOVE

By now, you will have tested all of the right leg's servos, so these should be sound (see Stage 46 if this is not the case). If these servos move, but the hip servo does not, follow Steps 6-14 of this stage in reverse and check the servo's connectors and, if necessary, test/reset its ID number.



DISASSEMBLY

Follow the assembly of Steps 6-14 of this stage in reverse, and check the connections carefully. If this does not solve the issue, you will need to test/reset the hip servo. Its ID number should be 2, and more information

on the process of resetting this servo's ID number is found in its original stage, Stage 33.

COMING IN PACK 14

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 51

Prepare and fit Robi's left hip servo

PARTS PROVIDED

The servo that will move Robi's left hip.



Stage 52

Join Robi's legs for the first time

PARTS PROVIDED

More hip components and a soccer ball.



Stage 53

Start working on Robi's head and neck

PARTS PROVIDED

Neck components and another servo cable.



Stage 54

Prepare Robi's neck servo

PARTS PROVIDED

The servo that will turn Robi's head from side to side.